Su-won Jin

Introduction & Projects

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Introduction

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"I want to study robotic intelligence that understands human language and interacts with the real world"

Name

- Su-won Jin (진수원)

Education

- Korea Maritime and Ocean University,
- B.S. in Electrical and Electronic Engineering
- Total GPA: 4.0 / 4.5, Major GPA: 3.98 / 4.5

Experience

- Undergraduate Researcher / Jul. 2024 – Jan. 2025
  - Embedded Signal Processing Lab
- Course-based Intern / Jan. 2025 – Feb. 2025
  - SEANET, Engineering Design Team

Research Interests

- Vision-Language-Action Models
- Multi-Modal Learning
- Robot Perception
- Robot Task Planning

- I am interested in enabling robots to understand natural language instructions and visual information, ground target objects and spatial relationships in 3D scenes, and generate high-level task plans for executing robot actions.

VLA

Perception $\xrightarrow{A}$ **Grounding** $\xrightarrow{B}$ **Task Planning** $\xrightarrow{C}$ **Robot Actions**

VLA's Task Planner Question	Answer
1. Is the target object localized?	Yes
2. Is the target object reachable?	Yes
3. Is the target object graspable?	Yes
4. Is the target location identified?	Yes

<Subtask Plans and Actions>

approach → grasp → move → place

Process A: Language-Grounded 3D Scene Understanding

1. Grounding language instructions and visual information
- Visual Input: ⓐ => glass, ⓑ => paper cup, ⓒ => sink
- Language Input: ⓒ put => place

Process B: Grounding-based Task Planning and Actions

2. Generate subtask plans from grounded scene infromation
- Target: glass
- Goal: sink
- Action sequence: approach → grasp → move → place

A BLE-Based Bus Boarding System for the Visually Impaired

A BLE-Based Bus Boarding System for the Visually Impaired

- **Barriers to Bus Use Among the Visually Impaired**

Reason	**Percentage (%)**
Difficulty identifying the desired bus number	**24.5**
Difficulty locating boarding and exit doors	**8.4**

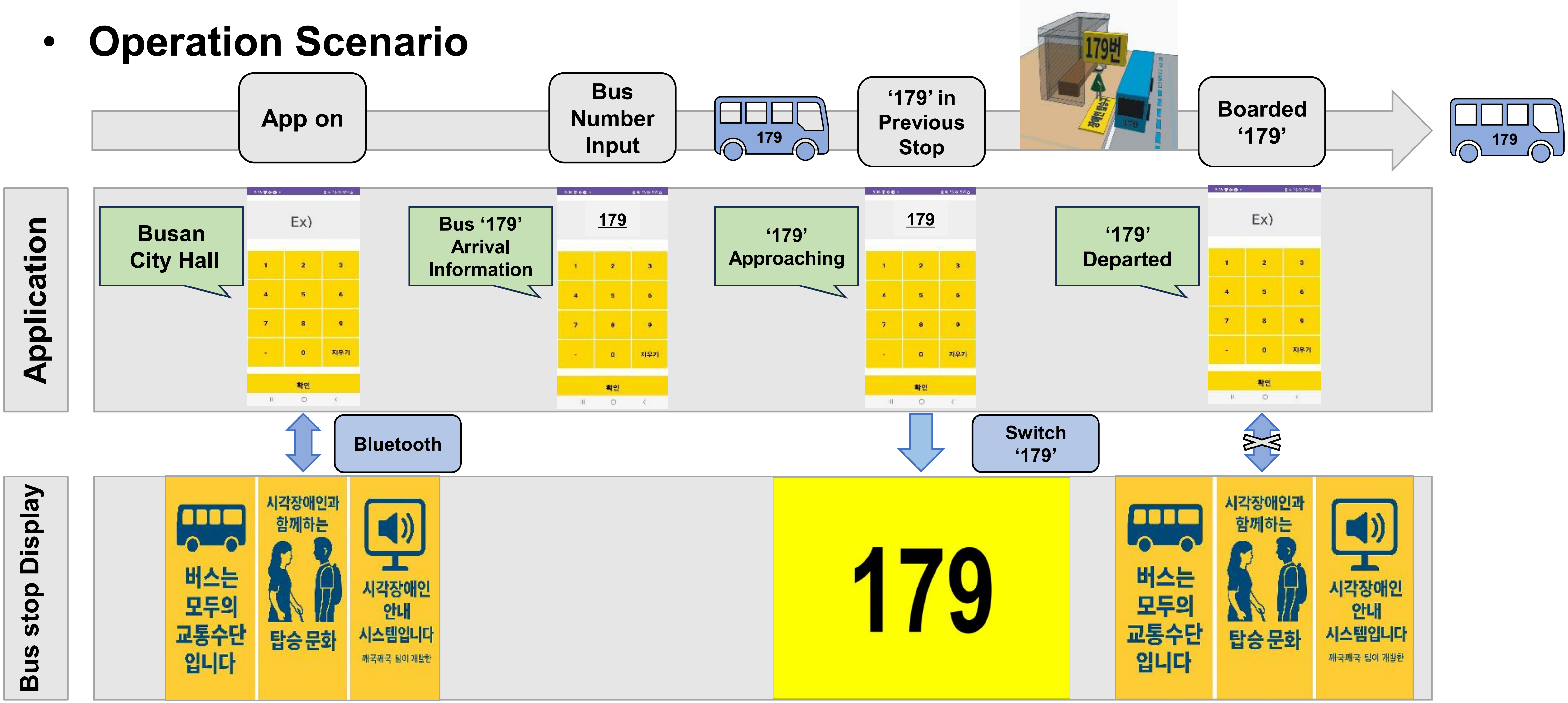
*Source: Wello, Korea Blind Union

- **Solutions**

  - ➢ Voice-guided application for bus number identification
  - ➢ Bus stop display with a designated boarding zone

A BLE-Based Bus Boarding System for the Visually Impaired

- **Operation Scenario**



< Demo Video >

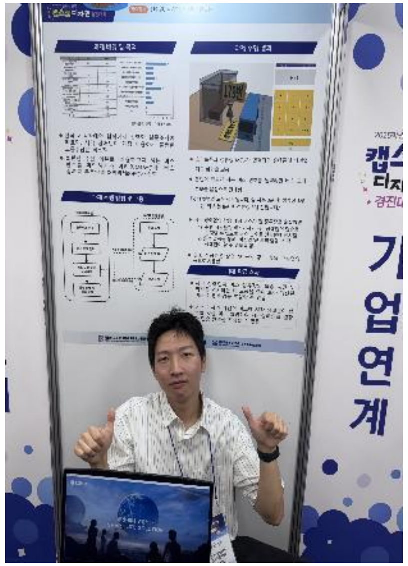
A BLE-Based Bus Boarding System for the Visually Impaired

- **Results**

2025.03 ~ 2025.06

  - BLE-based app-display bus boarding system
  - Competition & Conference participation
  - Gold prize in departmental competition

- **My Contribution**
  - OpenAPI integration & XML parsing
  - Event-driven display/alert logic
  - Team Leader
  - Paper preparation & submission
  - Poster presentations at competitions and a conference



< Competition >

시각 장애인을 위한 BLE 기반의 버스 승차 시스템

A BLE-Based Bus Boarding System for the Visually Impaired

< Paper >

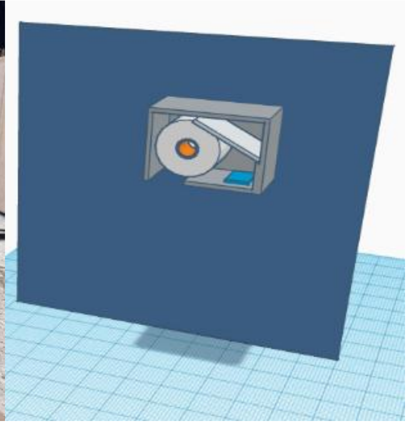
Real-time Toilet Paper Remaining Level Monitoring System

Real-time Toilet Paper Remaining Level Monitoring System

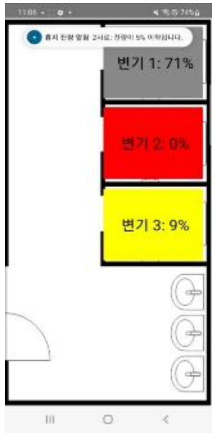
- **Problems**
  - ➢ Toilet paper availability is critical in high-traffic public restrooms
  - ➢ Manual inspection makes efficient restroom management difficult in multi-stall environments

- **Solutions**

< Dispenser >



< Installation Concept >

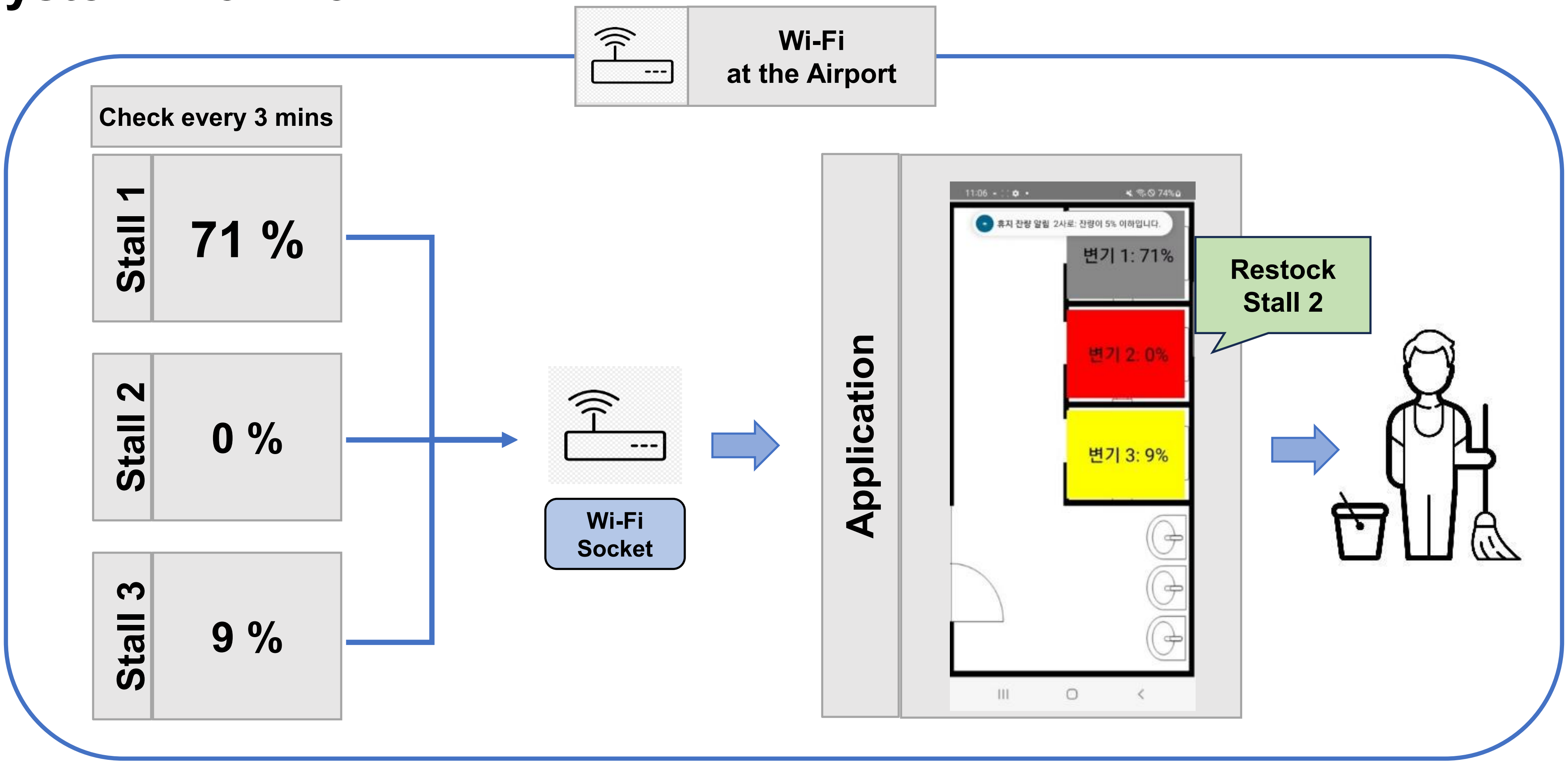


< Application >

- ➢ Arduino-based smart toilet paper dispenser for level detection
- ➢ Wi-Fi-based monitoring and management application

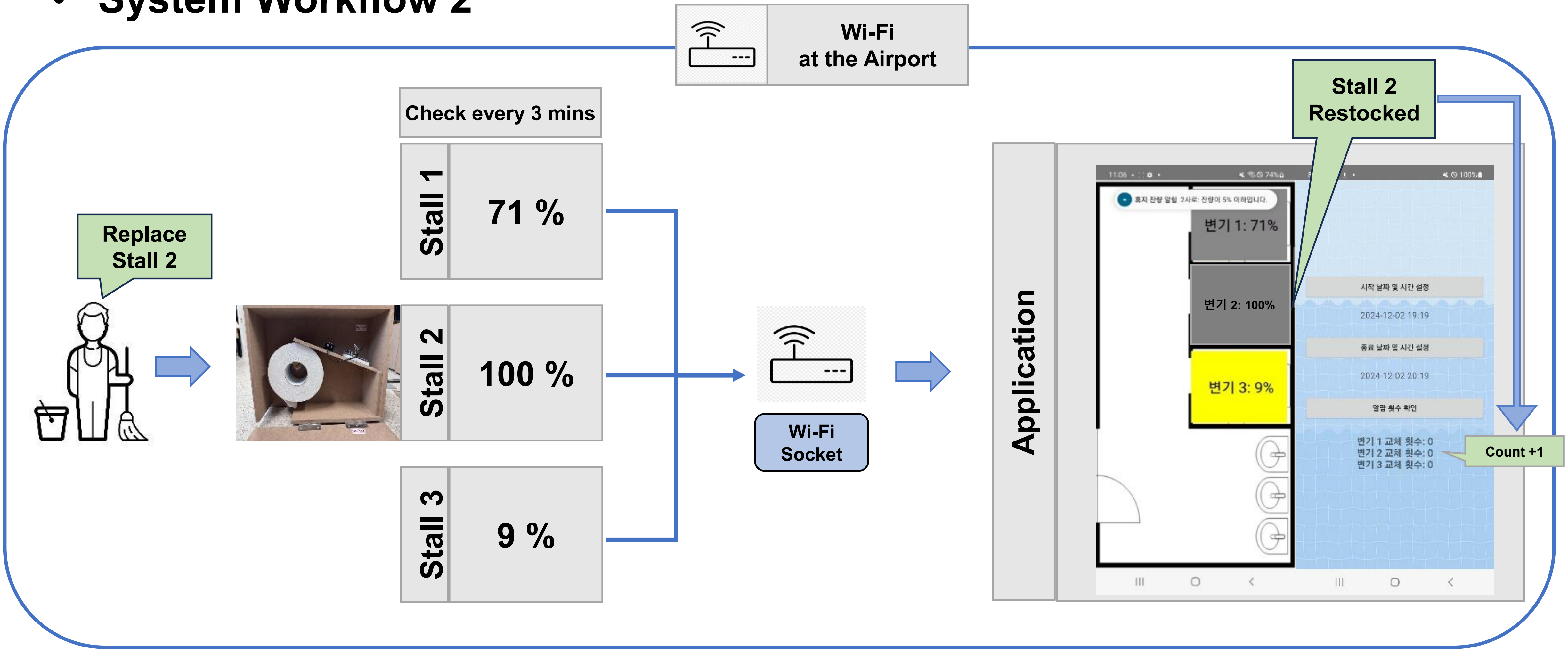
Real-time Toilet Paper Remaining Level Monitoring System

- **System Workflow 1**



Real-time Toilet Paper Remaining Level Monitoring System

- **System Workflow 2**



Real-time Toilet Paper Remaining Level Monitoring System

- **Results** — **2024.09 ~ 2024.12**
  - ➢ IMU-based toilet paper level detection dispenser
  - ➢ Real-time monitoring application
  - ➢ Competition awards & Patent pending

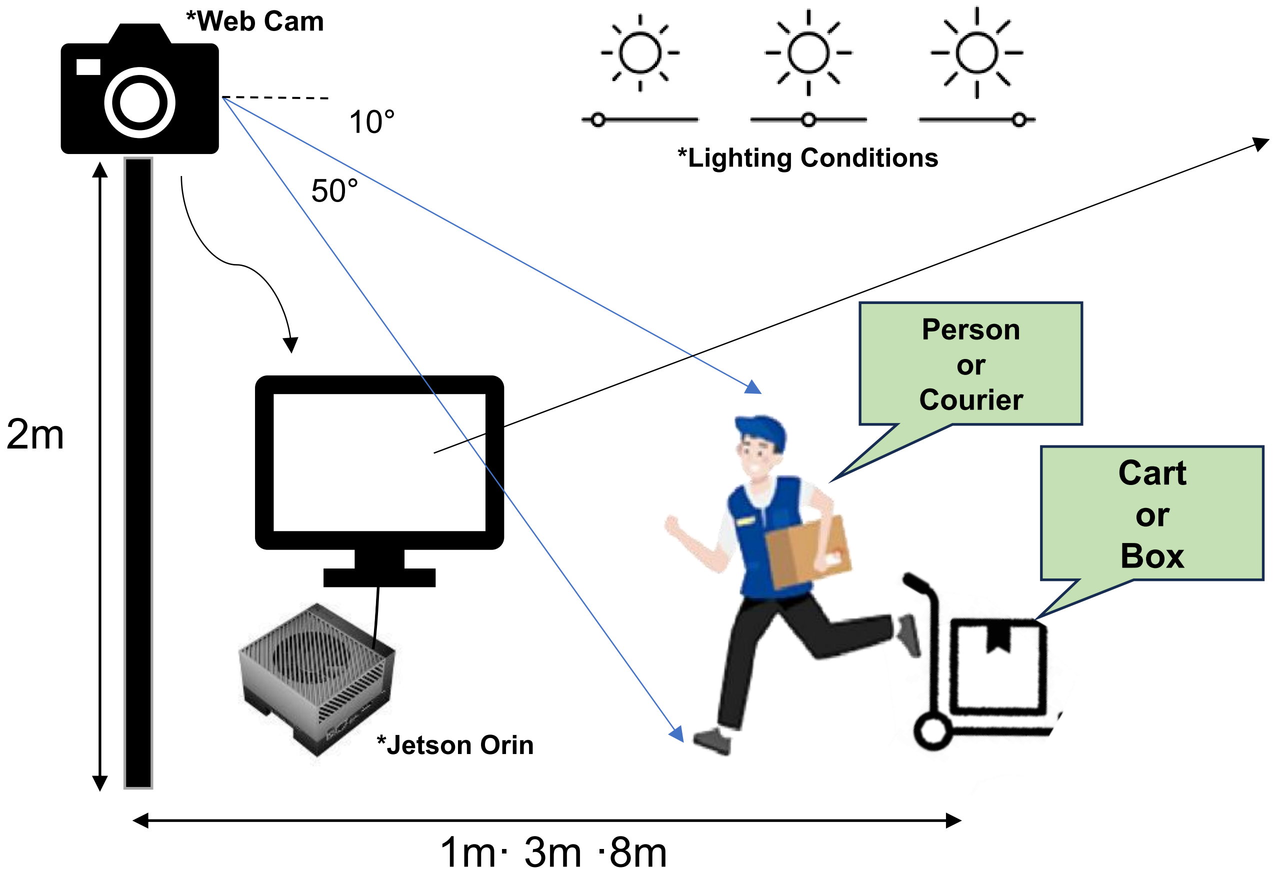
< Competition >

- **My Contribution**
  - ➢ Application UI development & sensor data visualization
  - ➢ IMU sensor data processing logic
  - ➢ Replacement event detection
  - ➢ Team Leader
  - ➢ Poster presentations at competitions

Development of High-Density Smart Parcel Terminal Storage Infrastructure and Management Technology

Development of High-Density Smart Parcel Terminal Storage Infrastructure and Management Technology

- **Experimental Overview:** Development of an object recognition system for an unmanned parcel locker



[timestamp] A person is approaching
[timestamp] A courier is approaching with
(cart or box)
⋮

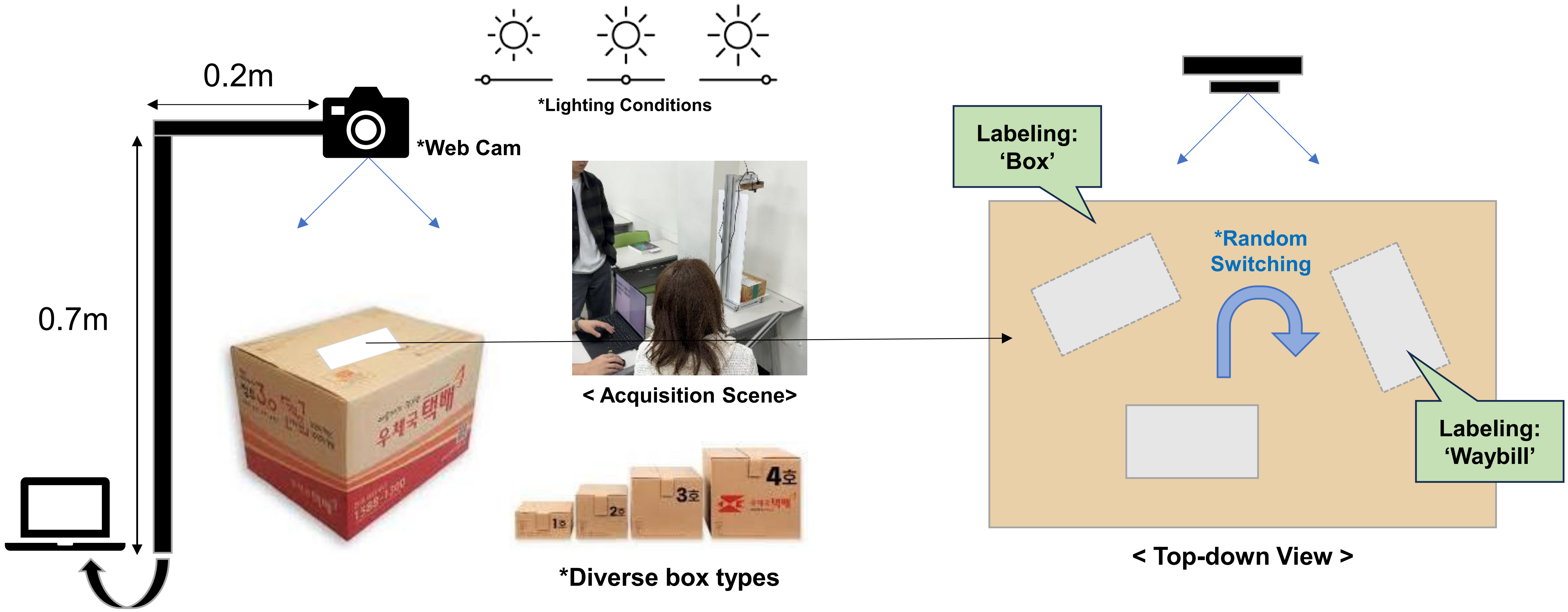
< Experimental Setup>

< Target Object >

< YOLO-based target object classification and predefined message output >

Development of High-Density Smart Parcel Terminal Storage Infrastructure and Management Technology

- **Dataset Acquisition & Labeling**



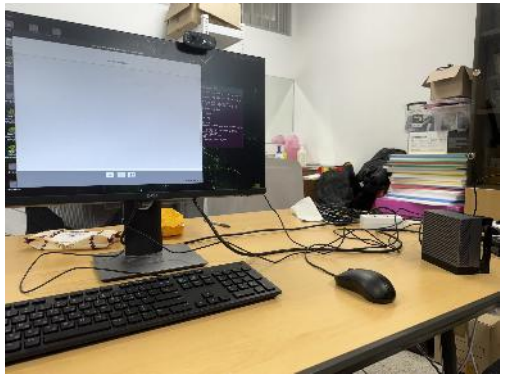
< Dataset with diverse lighting, box types, and waybill positions >

Development of High-Density Smart Parcel Terminal Storage Infrastructure and Management Technology

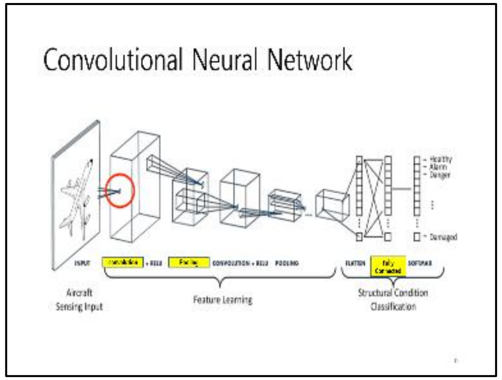
- # My Contribution

2024.07 ~ 2024.12

- **Testbed construction for object recognition model execution**
  - UI, Initial settings, Jetson Orin deployment
- **Validation environment design**
  - camera height, lighting, clothing, angle, and object conditions
- **Top-down dataset acquisition & annotation**
  - diverse lighting, box, and waybill positions
- **Fundamental study of neural networks and evaluation metrics**
  - Seminar presentation on CNN/RNN models and captioning metrics



< Jetson Orin Deployment >



< Seminar PPT >

< Click to view on [GitHub] >

Inter – University B.SORI Project Competition

Inter –University B.SORI Project Competition

- **Background**

  - ➢ Inter-university idea competition on local community issues
  - ➢ Two-day team formation and idea generation
  - ➢ Ten-day idea refinement and PPT presentation

- **Objective: Motorcycle noise reduction for different residential environments**

< Hillside Residential Area >

- **Paved detour road installation**

➢ Uphill motorcycle noise in hillside area
➢ Detour-based noise reduction strategy

< Urban Area >

- **Rear-view noise enforcement camera**

➢ Enhanced nighttime noise enforcement

Inter –University B.SORI Project Competition

2024.09 ~ 2024.09

- **Results**
  - ➢ Local issue analysis and solution ideation
  - ➢ Noise reduction ideas for residential environments
  - ➢ Concept proposal PPT
  - ➢ Encouragement award

< Initial Concept Sketch >

- **My Contribution**
  - ➢ Project direction and team coordination
  - ➢ Noise issue and regional characteristics analysis
  - ➢ Team Leader
  - ➢ Online PPT presentation

< Concept Proposal PPT >

< Click to view on [GitHub]>

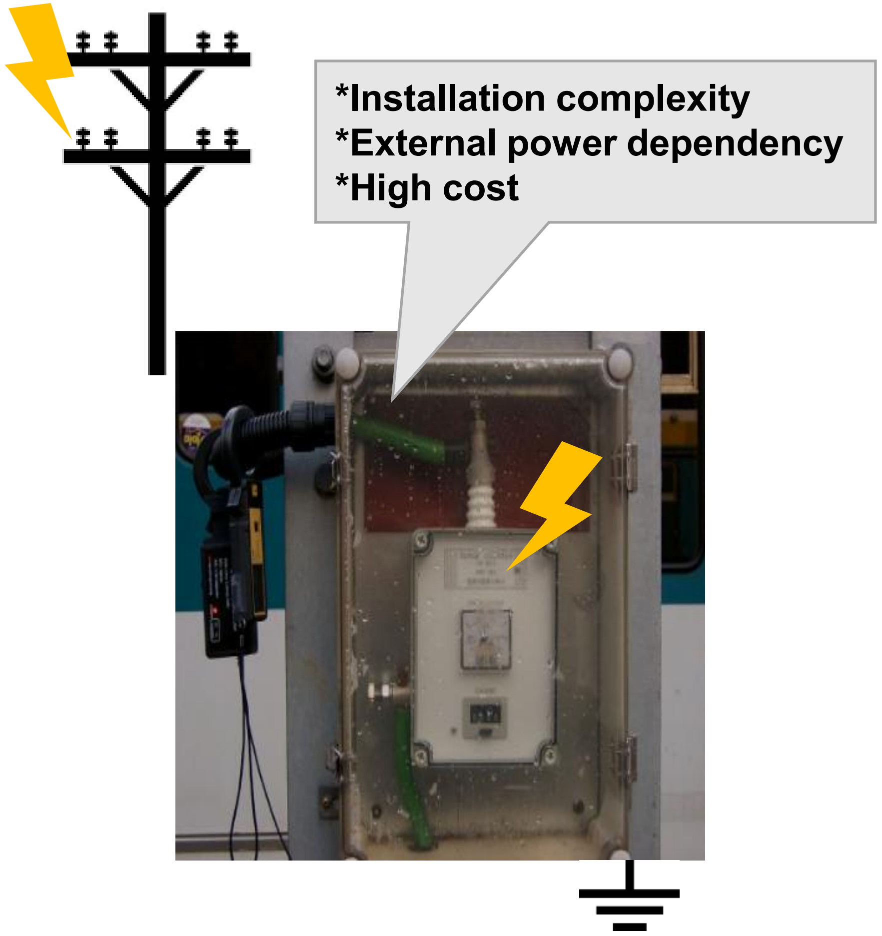
Proposal for a Non-Contact Surge(Lightning) Counter

Proposal for a Non-Contact Surge(Lightning) Counter

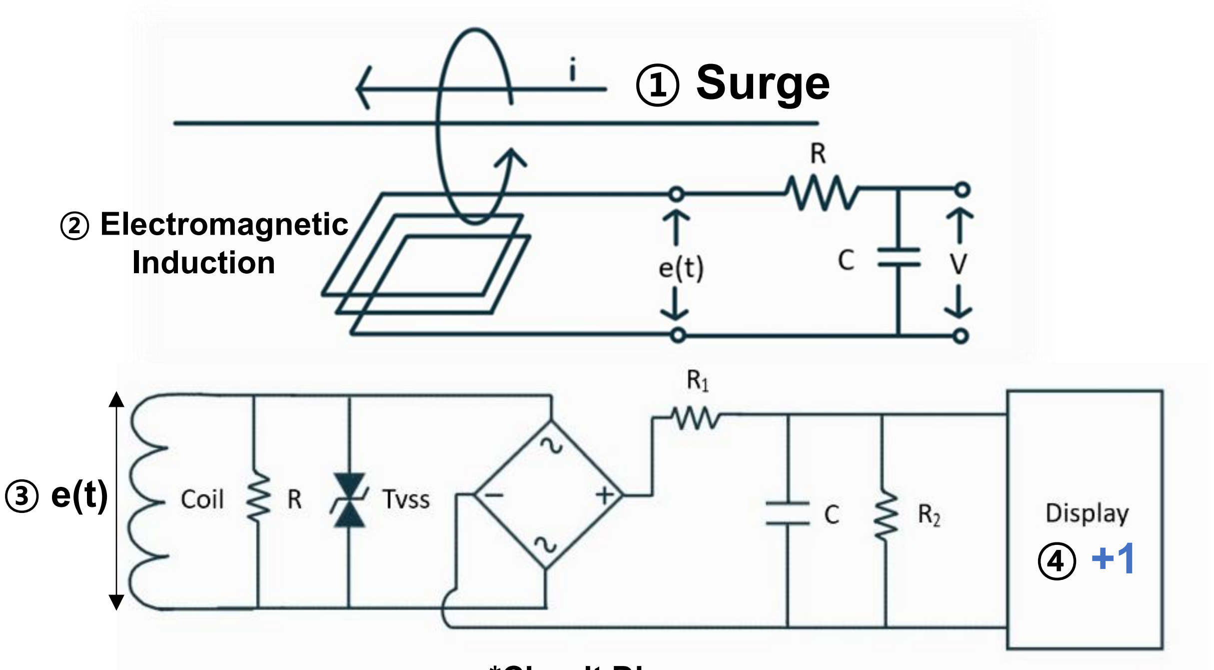
- **Background**
  - ➢ Student railway idea competition hosted by the Korean Society for Railway
  - ➢ Team-based railway idea proposal in PPT format
  - ➢ Participation in a non-contact surge (lightning) counter proposal under faculty guidance
- **Objective**
  - ➢ Proposal of a non-contact surge counter using electromagnetic induction without external power
  - ➢ Improvement of installation complexity, external power dependency, and high cost in conventional equipment

< Proposal Counter >

Proposal for a Non-Contact Surge(Lightning) Counter



< Conventional Counter >



*Circuit Diagram

< Installation Example >

Proposal for a Non-Contact Surge(Lightning) Counter

2024.05 ~ 2024.06

- **Results**
  - ➢ Non-contact surge counter concept proposal
  - ➢ Improvement measures over conventional equipment
  - ➢ Concept proposal PPT

< Concept Proposal PPT >

< Click to view on GitHub >

- **My Contribution**
  - ➢ Study of surge counters, surge, electromagnetic induction, and circuit configurations
  - ➢ Analysis of conventional surge counter structures and conceptualization of a non-contact approach
  - ➢ Team Leader
  - ➢ PPT preparation and incorporation of faculty feedback